

TWO NONSEPARABLE $\mathbb{G}_0$ DICHOTOMIES

B. MILLER

ABSTRACT. We give classical proofs of a generalization of the uniform $\mathbb{G}_0$ dichotomy to families of digraphs indexed by elements of $\kappa^{\mathbb{N}}$ and Kanovei's κ -Souslin generalization of the $\mathbb{G}_0$ dichotomy.

INTRODUCTION

Suppose that κ is an infinite cardinal and X is a Hausdorff space. Endow κ with the discrete topology and $\kappa^{\mathbb{N}}$ with the product topology. For all $n \in \mathbb{N}$ and $s \in \kappa^n$, set $|s| = n$. For all $\sigma \in \kappa^{\leq \mathbb{N}}$, let $s \sqsubseteq \sigma$ indicate that $\forall i < |s| \ s(i) = \sigma(i)$. Define $\mathcal{N}_s = \{\sigma \in \kappa^{\mathbb{N}} \mid s \sqsubseteq \sigma\}$. We write $\forall^* \sigma \in \kappa^{\mathbb{N}} \ \phi(\sigma)$ to indicate that $\{\sigma \in \kappa^{\mathbb{N}} \mid \phi(\sigma)\}$ is comeager. The *projection* function from $\prod_{i \in I} X_i$ to X_i is given by $\text{proj}_{X_i}(x) = x(i)$. The *horizontal and vertical sections* of a set $R \subseteq X \times Y$ are given by $R^y = \{x \in X \mid x R y\}$ and $R_x = \{y \in Y \mid x R y\}$. Given a function $\pi: R \subseteq X \times Y \rightarrow Z$, define $\pi_x: R_x \rightarrow Z$ by $\pi_x(y) = \pi(x, y)$.

We say that $\phi: C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \rightarrow \kappa^{\mathbb{N}} \times X$ is *parameter preserving* if $\text{proj}_{\kappa^{\mathbb{N}}} \circ \phi = \text{proj}_{\kappa^{\mathbb{N}}} \upharpoonright C$, and $A \subseteq \kappa^{\mathbb{N}} \times X$ has *uniformly analytic (vertical) sections* if there are a closed set $C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and a continuous parameter-preserving function $\phi: C \rightarrow \kappa^{\mathbb{N}} \times X$ such that $A = \phi(C)$.

Recall that a sequence $(X_i)_{i \in I}$ of subsets of a topological space is *open separated* if there are pairwise disjoint open sets U_i such that $X_i \subseteq U_i$ for all $i \in I$. A subset of a topological space is *quasi-Borel* if it is in the closure of the open sets under complements, countable unions, and open-separated unions. A function $c: \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$ is *quasi-Borel* if $c^{-1}(\{n\})$ is quasi-Borel for all $n \in \mathbb{N}$.

A digraph on X is an irreflexive set $G \subseteq X \times X$. A *homomorphism* from a digraph G on X to a digraph H on Y is a function $\pi: X \rightarrow Y$ such that $(\pi \times \pi)(G) \subseteq H$. A *Y -coloring* of G is a homomorphism $c: X \rightarrow Y$ from G to inequality. Fix $\mathfrak{s} \in \prod_{n \in \mathbb{N}} 2^n$ such that $\mathfrak{s}(\mathbb{N})$ is dense in $2^{<\mathbb{N}}$ and define $\mathbb{G}_0 = \bigcup_{n \in \mathbb{N}} \{(\mathfrak{s}_n \cap (i) \cap c)_{i < 2} \mid c \in 2^{\mathbb{N}}\}$.

The uniform $\mathbb{G}_0$ dichotomy for digraphs indexed by $\mathbb{N}^{\mathbb{N}}$ is a straightforward corollary of the $\mathbb{G}_0$ dichotomy itself (see [Mil12, Theorem 31]). Here we show that the following generalization to digraphs indexed by $\kappa^{\mathbb{N}}$ follows from the classical proof of the $\mathbb{G}_0$ dichotomy and the

observation that the terminal independent sets arising in the classical proof can be grouped into countably many open-separated families, preserving both definability and the number of colors:

Theorem 1. *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ has uniformly analytic sections that are digraphs on X . Then exactly one of the following holds:*

- (1) *There is a quasi-Borel function $c: \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$ such that c_σ is a coloring of G_σ for all $\sigma \in \kappa^{\mathbb{N}}$.*
- (2) *There is a continuous homomorphism $\pi: 2^{\mathbb{N}} \rightarrow X$ from $\mathbb{G}_0$ to G_σ for some $\sigma \in \kappa^{\mathbb{N}}$.*

A subset of a topological space is κ -Souslin if it is a continuous image of a closed subset of $\kappa^{\mathbb{N}}$. A subset of a topological space is κ^+ -Borel if it belongs to the smallest family containing the open sets and closed under complements and unions of cardinality at most κ . A function $c: X \rightarrow \kappa$ is κ^+ -Borel if $c^{-1}(\{\alpha\})$ is κ^+ -Borel for all $\alpha < \kappa$.

By combining Theorem 1 with a Baire category argument, we obtain a classical proof of the following generalization of the $\mathbb{G}_0$ dichotomy:

Theorem 2 (Kanovei [Kan97]). *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and G is a κ -Souslin digraph on X . Then at least one of the following holds:*

- (1) *There is a κ^+ -Borel coloring $c: X \rightarrow \kappa$ of G .*
- (2) *There is a continuous homomorphism from $\mathbb{G}_0$ to G .*

In §1, we note closure properties of sets with uniformly analytic sections. In §2, we show that the horizontal sections of quasi-Borel sets have the Baire property. In §3, we establish a separation theorem for sets with uniformly analytic sections. In §4, we prove Theorem 1. And in §5, we prove Theorem 2.

1. POINTCLASSES

We direct the reader to [Kec95, §14] for the most basic facts concerning analytic sets. The following closure properties can be proven in the same fashion as the corresponding facts for analytic sets:

Proposition 1.1. *Suppose that κ is an aleph and X is a Hausdorff space. Then the subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections are closed under countable intersections and countable unions.*

Proposition 1.2. *Suppose that κ is an aleph, X and Y are Hausdorff spaces, $A \subseteq \kappa^{\mathbb{N}} \times X$ and $B \subseteq \kappa^{\mathbb{N}} \times Y$ have uniformly analytic sections, and $f: A \rightarrow \kappa^{\mathbb{N}} \times Y$ is continuous and parameter preserving. Then $f(A)$ and $f^{-1}(B)$ have uniformly analytic sections.*

Proposition 1.3. *Suppose that κ is an aleph, X is a Hausdorff space, and $A_s \subseteq \mathcal{N}_s \times X$ has uniformly analytic sections for all $s \in \kappa^{<\mathbb{N}}$. Then $\bigcup_{s \in \kappa^{<\mathbb{N}}} A_s$ has uniformly analytic sections.*

Proof. Fix closed sets $C_s \subseteq \mathcal{N}_s \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_s: C_s \rightarrow \kappa^{\mathbb{N}} \times X$ such that $A_s = \phi_s(C_s)$. If $n \in \mathbb{N}$, then $C_n = \bigcup_{s \in \kappa^n} C_s$ is closed and $\phi_n = \bigcup_{s \in \kappa^n} \phi_s$ is continuous, so $\bigcup_{n \in \mathbb{N}} \phi_n(C_n)$ has uniformly analytic sections by Proposition 1.1. $\square$

Proposition 1.4. *Suppose that κ is an aleph. Then every quasi-Borel subset of $\kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ has uniformly analytic sections.*

Proof. The closed subsets of $\kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ clearly have uniformly analytic sections. As metrizability ensures that the open sets are countable unions of closed sets, Proposition 1.1 ensures that every open subset of $\kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ has uniformly analytic sections.

By Proposition 1.1 and induction, it is sufficient to show that if $(U_i)_{i \in I}$ is a pairwise disjoint sequence of open subsets of $\kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and $B_i \subseteq U_i$ has uniformly analytic sections for all $i \in I$, then so too does $B = \bigcup_{i \in I} B_i$, as this implies that if $U_i \setminus B_i$ has uniformly analytic sections for all $i \in I$, then so too does $\sim B = \sim \bigcup_{i \in I} U_i \cup \bigcup_{i \in I} U_i \setminus B_i$.

For all $t \in \mathbb{N}^{<\mathbb{N}}$, note that the restriction of $\text{proj}_{\kappa^{<\mathbb{N}}}$ to $I_t = \{(i, s) \in I \times \kappa^{<\mathbb{N}} \mid \mathcal{N}_s \times \mathcal{N}_t \subseteq U_i\}$ is injective. Propositions 1.1 and 1.3 ensure that $B = \bigcup_{t \in \mathbb{N}^{<\mathbb{N}}} \bigcup_{(i,s) \in I_t} B_i \cap (\mathcal{N}_s \times \mathcal{N}_t)$ has uniformly analytic sections. $\square$

Proposition 1.5. *Suppose that κ is an aleph, X is a Hausdorff space, and $A \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly analytic sections. Then every relatively quasi-Borel subset of A has uniformly analytic sections.*

Proof. Fix a closed set $C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and a continuous parameter-preserving function $\phi: C \rightarrow \kappa^{\mathbb{N}} \times X$ such that $A = \phi(C)$. If $B \subseteq A$ is relatively quasi-Borel, then $\phi^{-1}(B)$ is relatively quasi-Borel in C . Propositions 1.4 and 1.2 therefore ensure that $B = \phi(\phi^{-1}(B))$ has uniformly analytic sections. $\square$

2. BAIRE CATEGORY

We direct the reader to [Kec95, §8] for the basic facts concerning Baire category used below. Although $\kappa^{\mathbb{N}}$ is not separable when κ is uncountable, localization holds in every completely metrizable space.

Proposition 2.1. *Suppose that κ is an aleph, X is a topological space, and $B \subseteq \kappa^{\mathbb{N}} \times X$ is quasi-Borel. Then every horizontal section of B has the Baire property.*

Proof. As the horizontal sections of B are quasi-Borel, this follows from the fact that the family of subsets of $\kappa^{\mathbb{N}}$ with the Baire property contains the open sets and is closed under complements, countable unions, and open-separated unions. $\square$

3. SEPARATION

Recall that $B \subseteq X$ separates $A_0, A_1 \subseteq X$ if $A_0 \subseteq B$ and $A_1 \cap B = \emptyset$.

Proposition 3.1. *Suppose that κ is an aleph, X is a Hausdorff space, and $A_0, A_1 \subseteq \kappa^{\mathbb{N}} \times X$ are disjoint sets with uniformly analytic sections. Then they are separated by a quasi-Borel set.*

Proof. Fix closed sets $C_i \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_i: C_i \rightarrow \kappa^{\mathbb{N}} \times X$ such that $A_i = \phi_i(C_i)$, and define $A_{i,s,t} = \phi_i(C_i \cap (\mathcal{N}_s \times \mathcal{N}_t))$ for all $i < 2$, $s \in \kappa^{<\mathbb{N}}$, and $t \in \mathbb{N}^{<\mathbb{N}}$.

Following the usual convention, we identify $\bigcup_{n \in \mathbb{N}} \kappa^n \times \mathbb{N}^n \times \mathbb{N}^n$ with $(\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$. Set $T = \{(s, t, u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}} \mid A_{0,s,t} \cap \overline{A_{1,s,u}} \neq \emptyset\}$.

Lemma 3.2. *The tree T is well-founded.*

Proof. Suppose, towards a contradiction, that $(\sigma, c_0, c_1) \in [T]$. If $i < 2$, then $(\sigma, c_i) \in C_i$, so we can set $x_i = (\text{proj}_X \circ \phi_i)(\sigma, c_i)$. As A_0 and A_1 are disjoint, it follows that $x_0 \neq x_1$, so there are disjoint open neighborhoods U_0 and U_1 of x_0 and x_1 . Fix $n \in \mathbb{N}$ such that $A_{i,\sigma \upharpoonright n, c_i \upharpoonright n} \subseteq \kappa^{\mathbb{N}} \times U_i$ for all $i < 2$. Then $A_{0,\sigma \upharpoonright n, c_0 \upharpoonright n} \cap \overline{A_{1,\sigma \upharpoonright n, c_1 \upharpoonright n}} = \emptyset$, which contradicts the fact that $(\sigma \upharpoonright n, c_0 \upharpoonright n, c_1 \upharpoonright n) \in T$. $\square$

Recursively define $B_{s,t,u} \subseteq \kappa^{\mathbb{N}} \times X$ for $(s, t, u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$ by

$$B_{s,t,u} = \begin{cases} \bigcup_{r \in S_s} (\mathcal{N}_r \times X) \cap \bigcup_{m \in \mathbb{N}} \bigcap_{n \in \mathbb{N}} B_{r,t \smallfrown (m), u \smallfrown (n)} & \text{if } (s, t, u) \in T, \\ \sim A_{1,s,u} & \text{otherwise.} \end{cases}$$

where $S_s = \{s \smallfrown (\alpha) \mid \alpha < \kappa\}$. The union in the first case is open-separated, since the sets $\mathcal{N}_r \times X$, for $r \in S_s$, are pairwise disjoint and open. The obvious induction therefore shows that $B_{s,t,u}$ is quasi-Borel and separates $A_{0,s,t}$ from $A_{1,s,u}$, so $B_{\emptyset, \emptyset, \emptyset}$ separates A_0 from A_1 . $\square$

Given $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$, we say that A is G -independent if $\forall \sigma \in \kappa^{\mathbb{N}} (A_\sigma \times A_\sigma) \cap G_\sigma = \emptyset$.

Proposition 3.3. *Suppose that κ is an aleph, X is a Hausdorff space, the sets $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ have uniformly analytic sections, and A is G -independent. Then there is a quasi-Borel G -independent set $B \supseteq A$.*

Proof. By the proof of [dRM21, Proposition 1.2], using Proposition 3.1 in place of Lusin's separation theorem. $\square$

4. THE UNIFORM $\mathbb{G}_0$ DICHOTOMY

The proof follows that of [dRM21, Theorem 1.4]. The principal difference is that terminality must be localized to basic open subsets of the parameter space. We therefore keep track of pairs consisting of an approximation and a finite parameter sequence, assemble the resulting independent sets by open-separated unions, and carry the parameter neighborhoods through the fusion.

Proof of Theorem 1. Define $p_i: \kappa^{\mathbb{N}} \times (X \times X) \rightarrow \kappa^{\mathbb{N}} \times X$ by setting $p_i(\sigma, (x_0, x_1)) = (\sigma, x_i)$ for all $i < 2$, $\sigma \in \kappa^{\mathbb{N}}$, and $x_0, x_1 \in X$. As Propositions 1.1 and 1.2 ensure that $\bigcup_{i < 2} p_i(G)$ has uniformly analytic sections, there are closed sets $C, D \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi: C \rightarrow \kappa^{\mathbb{N}} \times X$ and $\psi: D \rightarrow \kappa^{\mathbb{N}} \times (X \times X)$ such that $\bigcup_{i < 2} p_i(G) = \phi(C)$ and $G = \psi(D)$.

An *approximation* is a triple $a = (n^a, \phi^a, (\psi_k^a)_{k < n^a})$, where $n^a \in \mathbb{N}$, $\phi^a: 2^{n^a} \rightarrow \mathbb{N}^{\mathbb{N}}$, and $\psi_k^a: 2^{n^a - (k+1)} \rightarrow \mathbb{N}^{\mathbb{N}}$ for all $k < n^a$. An approximation b is a *one-step extension* of a if:

- (1) $n^b = n^a + 1$,
- (2) $\forall u \in 2^{n^a} \forall v \in 2^{n^b} (u \sqsubseteq v \implies \phi^a(u) \sqsubseteq \phi^b(v))$, and
- (3) $\forall k < n^a \forall u \in 2^{n^a - (k+1)} \forall v \in 2^{n^b - (k+1)} (u \sqsubseteq v \implies \psi_k^a(u) \sqsubseteq \psi_k^b(v))$.

Given $\sigma \in \kappa^{\mathbb{N}}$, a σ -*configuration* is a triple $\gamma = (n^\gamma, \phi^\gamma, (\psi_k^\gamma)_{k < n^\gamma})$, where $\phi^\gamma: 2^{n^\gamma} \rightarrow \mathbb{N}^{\mathbb{N}}$ and $\psi_k^\gamma: 2^{n^\gamma - (k+1)} \rightarrow \mathbb{N}^{\mathbb{N}}$ for all $k < n^\gamma$, such that:

- (1) $\forall u \in 2^{n^\gamma} \phi^\gamma(u) \in C_\sigma$,
- (2) $\forall k < n^\gamma \forall v \in 2^{n^\gamma - (k+1)} \psi_k^\gamma(v) \in D_\sigma$, and
- (3) $\forall k < n^\gamma \forall v \in 2^{n^\gamma - (k+1)}$
 $(\text{proj}_{X \times X} \circ \psi_\sigma \circ \psi_k^\gamma)(v) = ((\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(s_k \frown (i) \frown v))_{i < 2}$.

We say that γ is *compatible* with $Y \subseteq X$ if $(\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(2^{n^\gamma}) \subseteq Y$, and *compatible* with a if:

- (1) $n^\gamma = n^a$,
- (2) $\forall u \in 2^{n^a} \phi^a(u) \sqsubseteq \phi^\gamma(u)$, and
- (3) $\forall k < n^a \forall v \in 2^{n^a - (k+1)} \psi_k^a(v) \sqsubseteq \psi_k^\gamma(v)$.

Given a set $Z \subseteq \kappa^{\mathbb{N}} \times X$, we say that a is *Z-terminal* at σ if no σ -configuration is compatible with Z_σ and a one-step extension of a . Let $A(a, Z)$ denote the set of pairs of the form $(\sigma, (\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(s_{n^a}))$, where $\sigma \in \kappa^{\mathbb{N}}$ and γ is a σ -configuration compatible with a and Z_σ .

Lemma 4.1. *Suppose that a is an approximation, $\sigma \in \kappa^{\mathbb{N}}$, $Z \subseteq \kappa^{\mathbb{N}} \times X$, and a is Z -terminal at σ . Then $A(a, Z)_\sigma$ is G_σ -independent.*

Proof. See, for example, the proof of [dRM21, Lemma 1.5]. \(\boxtimes\)

Lemma 4.2. *If a is an approximation and $B \subseteq \kappa^{\mathbb{N}} \times X$ is quasi-Borel, then $A(a, B)$ has uniformly analytic sections.*

Proof. The set of pairs (σ, γ) for which γ is a σ -configuration compatible with a is closed. Compatibility with B_σ is a finite intersection of continuous preimages of B , and is therefore quasi-Borel. Proposition 1.5 and then Proposition 1.2 give the result. $\square$

Suppose that $U \subseteq \kappa^{\mathbb{N}}$. We say that a is Z -terminal on U if it is Z -terminal at σ for all $\sigma \in U$. Set $A_s(a, Z) = A(a, Z) \cap (\mathcal{N}_s \times X)$. If $B \subseteq \kappa^{\mathbb{N}} \times X$ is quasi-Borel and a is B -terminal on $\mathcal{N}_s$, then Proposition 3.3 and Lemmas 4.1 and 4.2 yield a G -independent quasi-Borel set $A_s(a, B) \subseteq B_s(a, B) \subseteq \mathcal{N}_s \times X$.

Let $\mathcal{A}$ denote the set of all approximations. We will recursively construct a decreasing sequence $(B^\alpha)_{\alpha < \kappa^+}$ of quasi-Borel subsets of $\kappa^{\mathbb{N}} \times X$ and an increasing sequence $(S^\alpha)_{\alpha < \kappa^+}$ of subsets of $\kappa^{<\mathbb{N}} \times \mathcal{A}$, starting with $S^0 = \emptyset$ and setting $S^\delta = \bigcup_{\alpha < \delta} S^\alpha$ for all limit ordinals $\delta < \kappa^+$. Suppose that $\alpha < \kappa^+$ and we have already constructed $(B^\beta)_{\beta < \alpha}$ and $(S^\beta)_{\beta \leq \alpha}$. Define $\rho^\alpha: S^\alpha \rightarrow \alpha$ by $\rho^\alpha(s, a) = \min\{\beta < \alpha \mid (s, a) \in S^{\beta+1}\}$, set $B^\alpha = \sim \bigcup_{n \in \mathbb{N}} \bigcup_{s \in \kappa^n} (\mathcal{N}_s \times X) \cap \bigcup_{(s, a) \in S^\alpha} B_s(a, B^{\rho^\alpha(s, a)})$ and $S^{\alpha+1} = S^\alpha \cup \{(s, a) \in \sim S^\alpha \mid a \text{ is } B^\alpha\text{-terminal on } \mathcal{N}_s\}$, and fix $B_s(a, B^\alpha)$ as above for all $(s, a) \in S^{\alpha+1} \setminus S^\alpha$. Each B^α is quasi-Borel, since the unions over a and n are countable and those over s are open separated. As $|\kappa^{<\mathbb{N}} \times \mathcal{A}| = \kappa$, there is a least $\lambda < \kappa^+$ with $S^\lambda = S^{\lambda+1}$, in which case $B^\lambda = B^{\lambda+1}$. Set $B = B^\lambda$, $S = S^\lambda$, and $\rho = \rho^\lambda$.

Lemma 4.3. *Suppose that $\alpha < \kappa^+$ and a is an approximation that is not $B^{\alpha+1}$ -terminal on $\mathcal{N}_s$ for some $s \in \kappa^{n^a}$. Then there is a one-step extension b of a that is not B^α -terminal on $\mathcal{N}_{s \smallfrown (\xi)}$ for some $\xi < \kappa$.*

Proof. Fix $\sigma \in \mathcal{N}_s$ for which a is not $B^{\alpha+1}$ -terminal at σ . Then there is a σ -configuration γ compatible with a one-step extension b of a and $B_\sigma^{\alpha+1}$. Set $\xi = \sigma(n^a)$. If b is B^α -terminal on $\mathcal{N}_{s \smallfrown (\xi)}$ and $\beta = \rho^{\alpha+1}(s \smallfrown (\xi), b)$, then $(\sigma, (\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(s_{n^b})) \in A_{s \smallfrown (\xi)}(b, B^\beta) \cap B^{\alpha+1} \subseteq B_{s \smallfrown (\xi)}(b, B^\beta) \cap B^{\alpha+1} = \emptyset$, the desired contradiction. $\square$

Let a_0 denote the unique approximation for which $n^{a_0} = 0$.

Lemma 4.4. *Suppose that a_0 is not B -terminal on $\kappa^{\mathbb{N}}$. Then there is a continuous homomorphism from $\mathbb{G}_0$ to G_σ for some $\sigma \in \kappa^{\mathbb{N}}$.*

Proof. By Lemma 4.3 there exist $a_n \in \mathcal{A}$ with $n^{a_n} = n$, $s_n \in \kappa^n$, $\sigma_n \in \mathcal{N}_{s_n}$, and a σ_n -configuration γ_n compatible with a_{n+1} and B_{σ_n} , where a_{n+1} is a one-step extension of a_n and $s_n \subseteq s_{n+1}$, for all $n \in \mathbb{N}$.

Set $\sigma = \bigcup_{n \in \mathbb{N}} s_n$. Then $\sigma_n \rightarrow \sigma$. Define $\phi': 2^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ and $\psi'_k: 2^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ by $\phi'(c) = \bigcup_{n \in \mathbb{N}} \phi^{a_n}(c \upharpoonright n)$ and $\psi'_k(c) = \bigcup_{n > k} \psi_k^{a_n}(c \upharpoonright (n - (k + 1)))$. If $c \in 2^{\mathbb{N}}$ and $k \in \mathbb{N}$, then $(\sigma_n, \phi^{a_n}(c \upharpoonright n + 1)) \rightarrow (\sigma, \phi'(c))$ and $(\sigma_n, \psi_k^{a_n}(c \upharpoonright (n - k))) \rightarrow (\sigma, \psi'_k(c))$, so the fact that C and D are closed ensures that $(\sigma, \phi'(c)) \in C$ and $(\sigma, \psi'_k(c)) \in D$.

Define $\pi: 2^{\mathbb{N}} \rightarrow X$ by $\pi(c) = (\text{proj}_X \circ \phi_\sigma)(\phi'(c))$ and observe that $(\text{proj}_{X \times X} \circ \psi_\sigma)(\psi'_k(c)) = (\pi(s_k \frown (i) \frown c))_{i < 2}$ for all $k \in \mathbb{N}$ and $c \in 2^{\mathbb{N}}$, so π is a continuous homomorphism from $\mathbb{G}_0$ to G_σ . $\square$

By Lemma 4.4, we can assume that a_0 is B -terminal on $\kappa^{\mathbb{N}}$. Set $\alpha = \rho(\emptyset, a_0)$. Then $A(a_0, B) = B \cap \bigcup_{i < 2} p_i(G)$ and $A(a_0, B^\alpha) \subseteq B_\emptyset(a_0, B^\alpha)$, so $A(a_0, B) = \emptyset$, thus B is G -independent. For all $a \in \mathcal{A}$ and $n \in \mathbb{N}$, define $S_{a,n} = \{s \in \kappa^n \mid (s, a) \in S\}$ and $B_{a,n} = \bigcup_{s \in S_{a,n}} B_s(a, B^{\rho(s,a)})$. Since the sets in this union are contained in the pairwise disjoint open sets $\mathcal{N}_s \times X$, the set $B_{a,n}$ is quasi-Borel. Then the latter sets are also G -independent, so (1) follows from the fact that $\mathcal{A}$ is countable and $\kappa^{\mathbb{N}} \times X = B \cup \bigcup_{a \in \mathcal{A}, n \in \mathbb{N}} B_{a,n}$.

It only remains to note that if both (1) and (2) hold, then there is a quasi-Borel $\mathbb{N}$ -coloring of $\mathbb{G}_0$. As $2^{\mathbb{N}}$ is second countable, every such function is Borel, contradicting [dRM21, Proposition 1.3]. $\square$

5. STRATIFICATIONS AND CATEGORY TRANSFORMS

For all sets $B \subseteq \kappa^{\mathbb{N}} \times X$ and sequences $t \in \kappa^{<\mathbb{N}}$, define

$$B^{*t} = \{x \in X \mid B^x \text{ is comeager in } \mathcal{N}_t\}.$$

Recall that a *network* for a space X is a family $\mathcal{F}$ of subsets of X such that every open subset of X is the union of sets in $\mathcal{F}$.

Proposition 5.1. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a topological space, $Y \subseteq X$ has a network of cardinality at most κ , and $B \subseteq \kappa^{\mathbb{N}} \times X$ is quasi-Borel. Then there is a κ^+ -Borel set $C \subseteq X$ such that $C \cap Y = B^{*t} \cap Y$.*

Proof. We argue by induction on the construction of B . If $B = U$ is open, set $U_v = \bigcup \{V \subseteq X \mid V \text{ is open and } \mathcal{N}_v \times V \subseteq U\}$. Then $U^{*t} = \bigcap_{u \sqsupseteq t} \bigcup_{v \sqsupseteq u} U_v$, which is κ^+ -Borel. For complements, localization gives $(\sim B)^{*t} = \sim \bigcup_{u \sqsupseteq t} B^{*u}$. For a countable or open-separated union $B = \bigcup_{i \in I} B_i$, localization gives $B^{*t} = \bigcap_{u \sqsupseteq t} \bigcup_{v \sqsupseteq u} \bigcup_{i \in I} B_i^{*v}$. In the open-separated case, this follows by localizing inside one of the separating open sets. After intersecting with $\kappa^{\mathbb{N}} \times Y$, only at most κ of the separating open sets are nonempty, since $\kappa^{\mathbb{N}} \times Y$ has a network of cardinality at most κ . The inductive hypothesis therefore yields the desired κ^+ -Borel representative in every case. $\square$

Given a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi: C \rightarrow X$, define $C' = \{(\sigma, b) \in \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \mid \sigma \circ b \in C\}$ and $\phi': C' \rightarrow \kappa^{\mathbb{N}} \times X$ by $\phi'(\sigma, b) = (\sigma, \phi(\sigma \circ b))$. A *stratification* of a κ -Souslin set $A \subseteq X$ is a set $A' \subseteq \kappa^{\mathbb{N}} \times X$ for which there exist a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi: C \rightarrow X$ such that $A = \phi(C)$ and $A' = \phi'(C')$.

Proposition 5.2. *Suppose that κ is an aleph, X is a Hausdorff space, $A \subseteq X$ is κ -Souslin, $A' \subseteq \kappa^{\mathbb{N}} \times X$ is a stratification of A , and $x \in A$. Then $\forall^* \sigma \in \kappa^{\mathbb{N}}$ $x \in A'_\sigma$.*

Proof. Fix a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi: C \rightarrow X$ with the property that $A = \phi(C)$ and $A' = \phi'(C')$, as well as $\tau \in C$ for which $\phi(\tau) = x$. Then the set $H = \bigcap_{n \in \mathbb{N}} \bigcup_{s \in \kappa^{< \mathbb{N}}} \mathcal{N}_{s \smallfrown (\tau(n))}$ is a dense G_δ subset of $\kappa^{\mathbb{N}}$ and $\forall \sigma \in H$ $x \in A'_\sigma$. $\square$

Proposition 5.3. *Suppose that κ is an aleph, $t \in \kappa^{< \mathbb{N}}$, X is a Hausdorff space, G is a κ -Souslin digraph on X , G' is a stratification of G , and $B \subseteq \kappa^{\mathbb{N}} \times X$ is a G' -independent quasi-Borel set. Then B^{*t} is G -independent.*

Proof. Otherwise fix $x, y \in B^{*t}$ with $x G y$ and appeal to Proposition 5.2 to obtain $\sigma \in \mathcal{N}_t$ with $x, y \in B_\sigma$ and $x G'_\sigma y$, a contradiction. $\square$

We can now establish the main result of this section:

Proof of Theorem 2. Fix a stratification G' of G . Set $V = \{x \in X \mid \exists y \in X (x G y \text{ or } y G x)\}$. As G is κ -Souslin, V has a network of cardinality at most κ . Then Theorem 1 allows us to assume there is a quasi-Borel function $c': \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$ such that c'_σ is a coloring of G'_σ for every $\sigma \in \kappa^{\mathbb{N}}$. Set $B_n = c'^{-1}(\{n\})$.

For all $n \in \mathbb{N}$ and $t \in \kappa^{< \mathbb{N}}$, Proposition 5.1 yields a κ^+ -Borel set $C_{n,t} \subseteq X$ such that $C_{n,t} \cap V = B_n^{*t} \cap V$. Proposition 5.3 ensures that B_n^{*t} is G -independent. Since every endpoint of an edge of G belongs to V , each $C_{n,t}$ is therefore G -independent.

If $x \in V$, then $\kappa^{\mathbb{N}} = \bigcup_{n \in \mathbb{N}} B_n^x$, so some B_n^x is non-meager. Proposition 2.1 and localization therefore give $t \in \kappa^{< \mathbb{N}}$ with $x \in B_n^{*t}$, hence $x \in C_{n,t}$. Thus $C_* = X \setminus \bigcup_{n \in \mathbb{N}, t \in \kappa^{< \mathbb{N}}} C_{n,t}$ is κ^+ -Borel and disjoint from V , and is therefore G -independent. The sets C_* and $C_{n,t}$ form a cover of X by at most κ many κ^+ -Borel G -independent sets, and hence yield a κ^+ -Borel coloring of G . $\square$

As noted in [Mil12, §8], the weakening of Theorem 2 without the definability condition on c follows from the simplification of the classical proof of the $\mathbb{G}_0$ dichotomy in which the use of separation is removed. As separation easily generalizes to κ -Souslin sets, this suggests a simpler

route towards a classical proof of the full version of Theorem 2. The direct adaptation of that proof, however, would require separation for continuous images of κ^+ -Borel sets. To see that this is consistently false, note that every Π_2^1 subset of $\mathbb{N}^{\mathbb{N}}$ is $\aleph_2$ -Borel (see, for example, [Kec95, Exercise 38.8]), so every Σ_3^1 subset of $\mathbb{N}^{\mathbb{N}}$ is a continuous image of an $\aleph_2$ -Borel set. Under $\text{ZFC} + \text{MA} + \neg\text{CH}$, the sets with the Baire property are closed under $\aleph_1$ -unions, so every $\aleph_2$ -Borel subset of $\mathbb{N}^{\mathbb{N}}$ has the Baire property. But under $\text{ZFC} + \text{MA} + \neg\text{CH} + \omega_1 = \omega_1^L$, there is a Δ_3^1 set $B \subseteq \mathbb{N}^{\mathbb{N}}$ without the Baire property by [HS85, Claim 8], so B is not $\aleph_2$ -Borel, but B and $\sim B$ are continuous images of $\aleph_2$ -Borel sets.

Acknowledgements. ChatGPT provided substantial assistance in the discovery and development of the results presented here.

REFERENCES

- [dRM21] N. de Rancourt and B. D. Miller, *The Feldman–Moore, Glimm–Effros, and Luzin–Novikov theorems over quotients*, 2021.
- [HS85] Leo A. Harrington and Saharon Shelah, *Some exact equiconsistency results in set theory*, Notre Dame J. Formal Logic **26** (1985), no. 2, 178–188.
- [Kan97] Vladimir Kanovei, *Two dichotomy theorems on colourability of non-analytic graphs*, Fundamenta Mathematicae **154** (1997), no. 2, 183–201.
- [Kec95] Alexander S. Kechris, *Classical descriptive set theory*, Graduate Texts in Mathematics, vol. 156, Springer-Verlag, New York, 1995.
- [Mil12] Benjamin D. Miller, *The graph-theoretic approach to descriptive set theory*, Bull. Symbolic Logic **18** (2012), no. 4, 554–575.